

Technical Documentation: Transport Layer Analysis

DEEP DIVE INTO TCP AND UDP MECHANISMS

1. Introduction to Layer 4 Protocols

The Transport Layer (Layer 4 of the OSI model) is responsible for end-to-end communication, session multiplexing, and process-to-process data delivery. The two primary protocols operating at this layer are the **Transmission Control Protocol (TCP)** and the **User Datagram Protocol (UDP)**.

As noted in the reference study logs, Layer 4 provides session multiplexing through the utilization of tracking identifiers known as **Port Numbers**. Source ports identify the local initiating session, while destination ports identify the specific Application Layer service intended to process the incoming payload.

Core Port Range Architectures:

- **Well-Known Ports: 0 — 1023** (Reserved for standardized core services like HTTP, HTTPS, SSH, and FTP).
- **Registered Ports: 1024 — 49151** (Assigned by IANA for specific vendor applications/ecosystems).
- **Ephemeral / Dynamic Private Ports: 49152 — 65535** (Allocated client-side dynamically to track unique outbound connections).

2. Transmission Control Protocol (TCP) Core Mechanisms

TCP is a connection-oriented, reliable transport protocol that guarantees ordered delivery and integrity checking through distinct operational mechanisms:

- **Reliable Communication & Acknowledgments:** Every transmitted segment expects a returning acknowledgment (**ACK**). If a packet is lost, the destination host will omit the corresponding acknowledgment, forcing a retransmission from the source.
- **Flow Control via Sliding Windows:** The receiver indicates its current buffer capacity using the **Window Size** parameter in the TCP header. This instructs the sender to dynamically increase or decrease the transmission volume before waiting for an explicit acknowledgment, preventing buffer saturation.
- **Data Sequencing:** Sequence numbers (**Seq**) ensure segments arriving out-of-order can be correctly reassembled by the receiving host.

2.1 The Connection Lifecycle

Establishing and tearing down connections involves structured cross-host handshakes:

Three-Way Handshake (Establishment)	Four-Step Disconnection (Teardown)
1. Client → Server: SYN 2. Server → Client: SYN-ACK 3. Client → Server: ACK	1. Initiator → Receiver: FIN 2. Receiver → Initiator: ACK 3. Receiver → Initiator: FIN 4. Initiator → Receiver: ACK

3. Wireshark Packet Capture Analysis (Simulated)

The following packet captures demonstrate the real-world hexadecimal and control flag tracking utilized during session tracking, handshakes, and transmission windows.

Capture 1: TCP Connection Establishment (Three-Way Handshake)

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.50	10.0.0.80	TCP	74	500 → 80 [SYN] Seq=10 Win=64240 Len=0
2	0.004120	10.0.0.80	192.168.1.50	TCP	74	80 → 500 [SYN, ACK] Seq=50 Ack=11 Win=65535 Len=0
3	0.004450	192.168.1.50	10.0.0.80	TCP	66	500 → 80 [ACK] Seq=11 Ack=51 Win=64240 Len=0

Transmission Control Protocol, Src Port: 500, Dst Port: 80
 Sequence Number: 10 (relative sequence number)
 Acknowledgment Number: 11 (relative ack number in Packet 2)
 Flags: 0x012 (SYN, ACK) -> [Present in Packet 2]
 Window: 65535

Capture 2: TCP Connection Teardown Sequence

No.	Time	Source	Destination	Protocol	Length	Info
14	12.450120	192.168.1.50	10.0.0.80	TCP	66	500 → 80 [FIN, ACK] Seq=11 Ack=51 Win=64240 Len=0
15	12.453210	10.0.0.80	192.168.1.50	TCP	66	80 → 500 [ACK] Seq=51 Ack=12 Win=65535 Len=0
16	12.453890	10.0.0.80	192.168.1.50	TCP	66	80 → 500 [FIN, ACK] Seq=51 Ack=12 Win=65535 Len=0
17	12.454110	192.168.1.50	10.0.0.80	TCP	66	500 → 80 [ACK] Seq=12 Ack=52 Win=64240 Len=0

4. Application Mapping Reference

Services are routed to specific target ports based on structural requirements for reliability (TCP) or raw performance overhead reduction (UDP).

TCP Specific Architecture	UDP Specific Architecture	Dual Protocol Capability (TCP & UDP)
FTP Data (20) / Control (21) File manipulation requiring zero corruption.	DHCP Server (67) / Client (68) Broadcast base dynamic network assignment.	DNS (53) Domain query resolving (UDP primarily; TCP for zone transfers).
SSH (22) Secure terminal access management.	TFTP (69) Lightweight file transfer over UDP.	
Telnet (23) Legacy cleartext terminal execution.	SNMP Agent (161) / Manager (162) Asynchronous infrastructure monitoring alerts.	
SMTP (25) Mail delivery server routing.	Syslog (514) Centralized structural infrastructure event logging.	
HTTP (80) / HTTPS (443) Web transport layers (Clear / Encrypted).		
POP3 (110) Client mail download pipeline management.		